The Climate Risk & Resilience Portal (ClimRR) Metadata and Data Dictionary

Climate Data Layer Reference Document

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Contents

Metadata

This document contains metadata for all of the climate variables available at launch of the *alpha* release of the Climate Risk and Resilience (ClimRR) portal. The links below direct to the REST service of the gridded data. Metadata, descriptions, and field names were last updated on 8/25/2023.

1. Temperature Minimum – Annual
2. Temperature Minimum – Seasonal
3. Temperature Maximum – Annual
4. Temperature Maximum – Seasonal
5. Precipitation – Annual Total
6. Precipitation None – Annual Average
7. Wind Speed – Annual Average
8. Cooling Degree Days – Annual Total
9. Heating Degree Days – Annual Total
10. Fire Weather Index – Averages
11. Fire Weather Index – FWI Classes
12. Heat Index – Summer Seasonal

Climate Scenarios

Climate scenarios are the set of conditions used to represent estimates of future greenhouse gas (GHG) concentrations in the atmosphere. Climate models then evaluate how these GHG concentrations affect future (projected) climate. The data in ClimRR include model results from two future climate scenarios, called Representative Concentration Pathways (RCPs):

- **RCP4.5**: in this scenario, human GHG emissions peak around 2040, then decline.
- **RCP8.5**: in this scenario, human GHG emissions continue to rise throughout the 21st century.

Each RCP is modeled over a **mid-century period** (2045 to 2054) and **end-of-century period** (2085 to 2094). A **historical period** (1995 to 2004) is also modeled using GHG concentrations during this period.

Downscaled Climate Models

A global climate model (GCM) is a complex mathematical representation of the major climate system components (atmosphere, land surface, ocean, and sea ice) and their interactions. These models project climatic conditions at frequent intervals over long periods of time (e.g., every 3 hours for the next 50-100 years), often with the purpose of evaluating how one or more GHG scenarios (such as RCP4.5 or RCP8.5) will impact future climate. Most GCMs project patterns at relatively coarse spatial resolutions, using grid cells ranging from 100km to 200km.

The climate data presented in this portal has been downscaled to a higher spatial resolution (12km) to fill a growing need for risk analysis and resilience planning at the local level. We use *dynamical downscaling*, which applies the outputs of a GCM as inputs to a separate, high-resolution regional climate model. Dynamical downscaling accounts for the physical processes and natural features of a region, as well as the complex interaction between these elements and global dynamics under a climate scenario.

Argonne's dynamical downscaling uses the Weather Research and Forecasting (WRF) model, which is a regional weather model for North America developed by the National Center for Atmospheric Research. Scientists at Argonne dynamically downscaled three different GCMs, including:

- **CCSM:** The Community Climate System Model (Version 4) is a coupled global climate model developed by the University Corporation for Atmospheric Research with funding from the National Science Foundation, the Department of Energy, and the National Aeronautics and Space Administration. It is comprised of atmospheric, land surface, ocean, and sea ice submodels that run simultaneously with a central coupler component.
- **GFDL:** The Geophysical Fluid Dynamics Laboratory at the National Oceanic and Atmospheric Administration developed the Earth System Model Version 2G (note: the general convention, which we use, is to use the Laboratory's abbreviation to identify this model). It includes an atmospheric circulation model and an oceanic circulation model, and takes into account land, sea ice, and iceberg dynamics.
- **HadGEM:** The United Kingdom's Met Office developed the Hadley Global Environment Model 2—Earth System. It is used for both operational weather forecasting and climate research, and includes coupled atmosphere-ocean analysis and an earth system component that includes dynamic vegetation, ocean biology, and atmospheric chemistry.

Argonne's began its dynamical downscaling by conducting a validation study, in which the regional model (WRF) was run using inputs from the GCMs over a historical period (1995-2004). This 'backcasting' allowed for an assessment of the downscaled model's ability to reproduce observed local climate trends. Once validated, Argonne then supplied each individual GCM's outputs (CCSM, GFDL, and HadGEM) for each climate scenario (mid-century RCP4.5, mid-century RCP8.5, end-of-century RCP4.5, and end-of-century RCP8.5) to the WRF regional model, producing three different downscaled projections of future climate conditions for each scenario, along with downscaled historical data for each GCM.

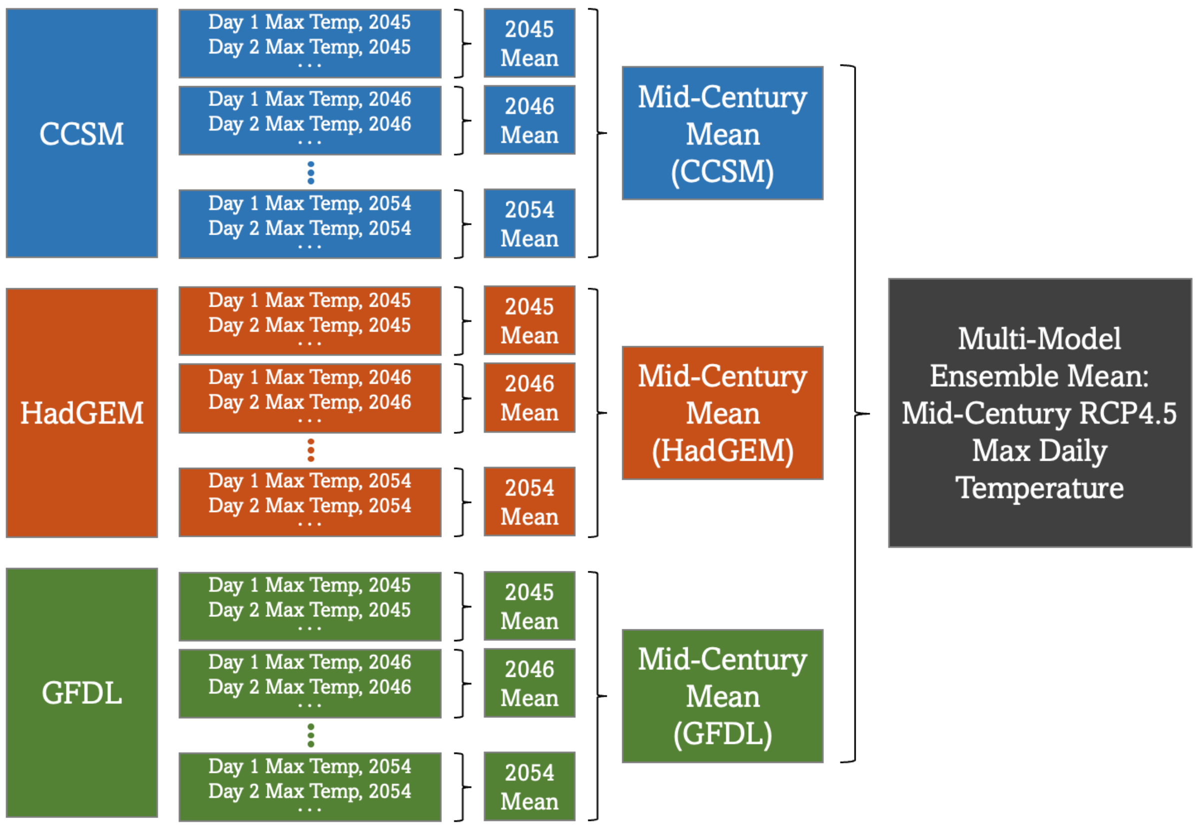
Ensemble Means

All data layers in ClimRR represent a climate variable along with its associated time period and climate scenario (e.g., mid-century RCP4.5). Each time period comprises one decade's worth of information: the **historical** (1995 – 2004), the **mid-century** (2045 – 2054), or the **end-of-century** (2085 – 2094). For each scenario, the WRF model is run with each of the three GCM outputs, producing three individual decades of weather data for each scenario. In other words, 30 years of downscaled climate data is produced for each decadal scenario. By using the outputs from three different GCMs, rather than a single model, Argonne's climate projections better account for the internal uncertainty associated with any single model.

Each year's worth of data includes weather outputs for every 3 hours, or 8 modeled outputs per day. While this allows for a high degree of granularity in assessing future climate trends, there are many different ways to analyze this data; however, there are several important common methodologies shared across all variables presented in this portal. Most variables are presented as annual or seasonal averages of daily observations, yet each annual/seasonal average draws upon all three different climate model runs for that scenario and the ten years of data produced by each model. Therefore, each variable (e.g., total annual precipitation) for a given scenario (e.g., mid-century RCP4.5) is produced by calculating an individual estimate for each of the 30 years of climate data associated with that scenario, and then taking the average of the 30 estimates. This result is what we term the **ensemble mean**.

An example illustration of ensemble mean calculations is provided in Figure 1. In this example, the annual average maximum daily temperature is being calculated for a single scenario: mid-century RCP4.5.

Figure 1: Illustration of ensemble mean calculations for maximum daily temperature at mid-century under RCP4.5



Temperature (Annual)

Each climate model generates temperature readings every 3 hours, or 8 temperature readings per day. The maximum daily temperature refers to the highest of these 8 readings, which often occurs in the middle of the daytime and is comparable to the 'high temperature' in a daily weather forecast. Similarly, the minimum daily temperature refers to the lowest of these 8 readings, which often occurs overnight and is comparable to the 'low temperature' in a daily weather forecast. Argonne calculated the annual average of both the maximum and minimum daily temperatures. These calculations involved extracting the highest temperature reading and lowest temperature reading for each individual day of a year (e.g., 2045) within a given scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM). These daily high/low readings were then used to calculate the annual average maximum or minimum daily temperature for that scenario's model year (e.g., the average max daily temperature for 2045 using the CCSM model under RCP4.5). This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) across all three climate models (CCSM, GFDL, and HadGEM). Finally, the 30 individual annual averages for a given time period/scenario were themselves averaged, producing a multi-model ensemble mean that represents the annual average of the maximum or minimum daily temperature for a given time period/scenario.

Temperature (Seasonal)

Each climate model generates temperature readings every 3 hours, or 8 temperature readings per day. The maximum daily temperature refers to the highest of these 8 readings, which often occurs in the middle of the daytime and is comparable to the 'high temperature' in a daily weather forecast. Similarly, the minimum daily temperature refers to the lowest of these 8 readings, which often occurs overnight and is comparable to the 'low temperature' in a daily weather forecast. Argonne calculated the seasonal average of both the maximum and minimum daily temperatures; the seasons are segmented as winter (Dec, Jan, Feb), spring (March, April, May), summer (June, July, Aug), and autumn (Sep, Oct, Nov). These calculations involved extracting the highest temperature reading and lowest temperature reading for each individual day of a year (e.g., 2045) within a given time period/scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM). These daily high/low readings were then classified by season and used to calculate the seasonal average maximum or minimum daily temperature for that scenario's model year (e.g., the average max daily temperature for summer of 2045 using the CCSM model under RCP4.5). This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) across all three climate models (CCSM, GFDL, and HadGEM). Finally, the 30 individual seasonal averages for a given time period/scenario were themselves averaged, producing a multi-model ensemble mean that represents the seasonal average of the maximum or minimum daily temperature for a given time period/scenario.

Precipitation

Each climate model estimates an amount of precipitation (whether rain, snow, sleet, or ice) that occurs every 3 hours across the entire modeled time period (i.e., every 3 hours, of every day, for all modeled years). These 3-hour precipitation estimates can be used to calculate the total precipitation over a designated period of time, ranging from daily to annually. Argonne calculated total annual precipitation by adding all 3-hour precipitation estimates for a given year (e.g., 2045) within a given time period/scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM), which produced the total annual precipitation for that scenario's model year, such as CCSM's estimate of annual precipitation in 2045 under climate scenario RCP4.5. This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) and across all three climate models (CCSM, GFDL, and HadGEM), producing a total of 30 estimates of total annual precipitation for a given time period/scenario. The average of these values was taken to produce the ensemble mean of the total annual precipitation (in inches) for each time period/scenario.

Wind Speed

Each climate model generates estimated wind speed readings (in miles per hour, or mph) every 3 hours, or 8 wind speed readings per day. These values are irrespective of wind direction and only represent the magnitude of wind speed. Using these readings, Argonne calculated the average daily wind speed (the average of each day's 8 wind speed readings) for every day within a given time period/scenario (e.g., mid-century RCP4.5) and for each climate model (CCSM, GFDL, and HadGEM). Argonne then took the average of each daily average wind speed for a given year (e.g., 2045) within a given time period/scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM), which produced the annual average wind speed for that scenario's model year, such as CCSM's estimate of annual average wind speeds in 2045 under climate scenario RCP4.5. This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) and across all three climate models (CCSM, GFDL, and HadGEM), producing a total of 30 estimates of annual average wind speeds for a given time period/scenario. The average of these values was taken to produce the ensemble mean of the annual average wind speed (in mph) for each time period/scenario.

Consecutive Days with No Precipitation

Each climate model estimates an amount of precipitation (whether rain, snow, sleet, or ice) that occurs every 3 hours across the entire modeled time period (i.e., every 3 hours, of every day, for all modeled years). These 3-hour precipitation estimates were used to calculate daily precipitation quantities by adding all 8 precipitation readings for each day of a given year (e.g., 2045) within a given time period/scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM). This process produced the total daily precipitation for every day in a scenario's model year, such as CCSM's daily estimates of total precipitation for the year 2045 under climate scenario RCP4.5. Using this information, Argonne identified the greatest number of *consecutive* days in which *no* precipitation occurred (i.e., the total daily precipitation quantity equaled zero) for that scenario's model year (e.g., for the year 2045 under scenario RCP4.5, the highest number of consecutive days without any precipitation was X). This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) and across all three climate models (CCSM, GFDL, and HadGEM) producing 10 yearly values for each model, with each value representing the longest consecutive span with no precipitation for that year. Of the 10 yearly values for each climate model, the maximum value was selected (e.g., the decadal maximum). This resulted in 3 values for the longest consecutive number of days without precipitation for each time period/scenario, with one value for each climate model’s 10 years of data. The average of these maximum of the maxima was then taken to produce the ensemble mean of the decade’s highest number of consecutive days without precipitation in a single year.

Degree Days (HDD/CDD)

Degree days measure how cold or warm a location is by comparing the daily average temperature to a reference temperature, usually 65°F (18.33°C). Heating or cooling degree days roughly correlate with building heating or cooling needs, providing a simple useful estimate in energy planning. Heating degree days (HDD) measure how cold the temperature was on a given day or period of days, which shapes the need for a certain amount of building heating. For example, a day with an average temperature of 40°F has 25 HDD (65 - 40 = 25 HDD). Similarly, cooling degree days (CDD) measure how hot the temperature was on a given day; a day with an average temperature of 90°F has 25 CDD (90 – 65 = 25 CDD).

The calculations of HDD and CDD for this portal began with each climate model, which generates temperature readings every 3 hours, or 8 temperature readings per day. For each, the maximum temperature (the daily high) and the minimum temperature (the daily low) were extracted and use to estimate the daily average temperature. This daily average temperature was then used to calculate that day’s HDD (if the average temperature was below 65°F) or CDD (if the average temperature was above 65°F). This process was conducted for each individual day of a year (e.g., 2045) within a given time period/scenario (e.g., mid-century RCP4.5) and for a given climate model (e.g., CCSM). These daily HDD/CDD values were then used to calculate the total number of HDD/CDD for that scenario's model year (e.g., the total HDD for 2045 using CCSM model under RCP4.5). This process was repeated for each year within a given time period/scenario (e.g., 2046, 2047, and so forth) across all three climate models (CCSM, GFDL, and HadGEM). Finally, the 30 annual total HDD/CDD estimates for a given time period/scenario were themselves averaged, producing a multi-model ensemble mean that represents the annual total HDD or CDD for a given time period/scenario.

Fire Weather Index

The Fire Weather Index (FWI), also known as the Canadian Forest Fire Weather Index, was developed by the Canadian Forest Service. It estimates wildfire danger using information on weather conditions that influence the spread of wildfires. The FWI is comprised of multiple components that are developed using daily readings of temperature, relative humidity, wind speed, and 24-hour precipitation. This weather information is used to estimate the dryness of soils and organic materials (which act as fuel for wildfires) and the rate of fire spread. The FWI is useful for evaluating weather-based conditions that heighten the danger of wildfire spread *once ignition has occurred*; it does not account for sources of ignition, which can have both natural and human causes.

The FWI ranges from zero to infinity, with higher numbers corresponding to greater fire danger. The level of wildfire danger, as represented by FWI, varies based on regional characteristics, such as a region's typical level of fire danger and its land cover. For example, areas in the U.S. Southwest, which are often exceptionally dry, will have higher average daily FWI values than areas in the Northeast. A relatively high FWI that suggests a heightened level of fire danger in the Northeast might correspond to average fire danger in the Southwest. Despite this variation, FWI values in excess of 20 typically represent high levels of fire danger, with levels above 30 representing very high to potentially extreme levels of fire danger. A representative example of fire danger classes used by the European Forest Fire Information System can be found here.

Components of the FWI

The FWI is comprised of six components that are developed using daily readings of temperature, relative humidity, wind speed, and 24-hour precipitation. These readings are taken at solar noon and reflect fire danger at early- to mid-afternoon, typically the time of day in which fire weather conditions are more conducive to fire spread.

Three of the FWI's components estimate the amount of moisture on and beneath the forest floor. These include measures of the moisture content of:

1. Fine fuels or forest litter (e.g., dried leaves) on top of the forest floor;
2. Intermediate organic layers, such as decomposing plant matter;
3. Deep organic layers and soils, which correspond to drought measures.

These moisture content components feed into the two primary subindices that generate the FWI. The first of these is the Initial Spread Index, which measures the expected rate of fire spread; it is based on wind speeds and the moisture content of fine fuels (#1 above). The second is the Buildup Index, which represents the total amount of forest fuel available for consumption, as measured via moisture in intermediate and deep organic layers (#2 and #3 above). The Initial Spread Index and the Buildup Index combine to generate the Fire Weather Index, or the index depicted in ClimRR. Figure 2, adapted from the Canadian Forestry Service, provides an illustration of the weather inputs, moisture codes, and sub-indices used to develop the final index (FWI) value.

For more information on the FWI and its methodology, please visit the National Wildfire Coordinating Group[1] or the Canadian Forestry Service's report[2] detailing FWI's development and structure.

Calculating Seasonal Averages

Daily FWI values were calculated using daily weather readings from Argonne's downscaled 12km climate data. For a detailed description of each weather variable included in the calculations, along with the formulas used to calculate FWI, see Van Wagner and Pickett.[3]

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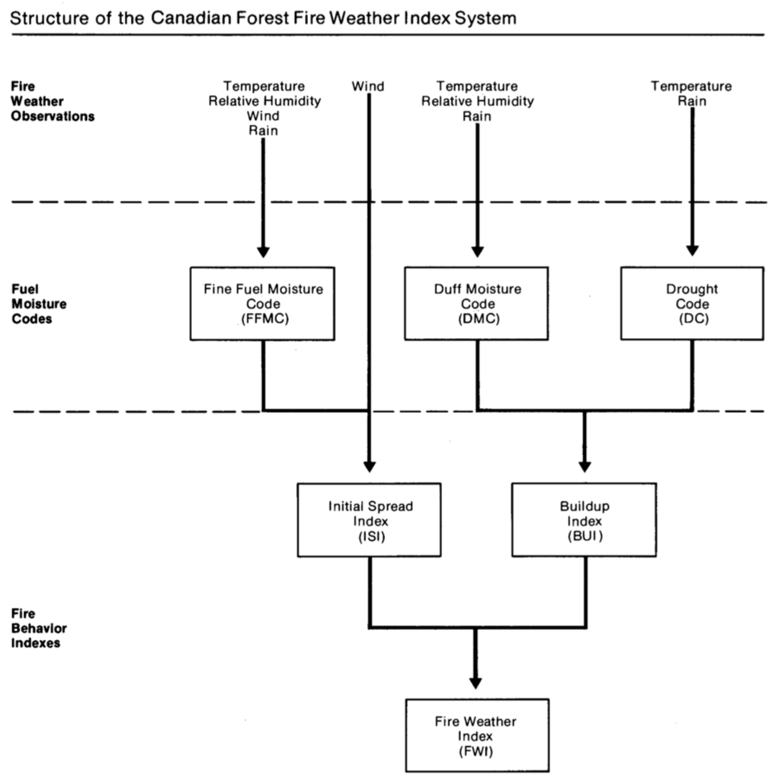
[1] National Wildfire Coordinating Group. (2023). Fire Weather Index (FWI) System. https://www.nwcg.gov/publications/pms437/cffdrs/fire-weather-index-system

[2] Van Wagner, C.E. (1987). *Development and Structure of the Canadian Forest Fire Weather Index System* (Forestry Technical Report 35). Canadian Forestry Service. https://cfs.nrcan.gc.ca/pubwarehouse/pdfs/19927.pdf

[3] Van Wagner, C.E., and T.L. Picket. (1985). *Equations and FORTRAN Program for the Canadian Forest Fire Weather Index System* (Forestry Technical Report 33). Canadian Forestry Service. https://cfs.nrcan.gc.ca/pubwarehouse/pdfs/19973.pdf

Argonne calculated the average daily FWI value by season for each year within a given scenario (e.g., mid-century RCP4.5) and climate model (e.g., CCSM). The seasons are segmented as winter (Dec, Jan, Feb), spring (March, April, May), summer (June, July, Aug), and autumn (Sep, Oct, Nov). Each year's seasonal average daily FWI was used to calculate a decadal average daily FWI value for that season (e.g., the average daily FWI value in winter at mid-century under RCP4.5). This process was repeated across all scenarios and all three climate models (CCSM, GFDL, and HadGEM), followed by calculating the multi-model ensemble mean of each scenario's seasonal average daily FWI value.

Figure 2: Structure of FWI inputs and components *(adapted from Canadian Forestry Service, 1984[4])*



Calculating FWI Classes

Daily FWI values were used to calculate seasonal and annual 95th percentile FWI values for all grid cells, across historical, mid-century RCP8.5, and end-of-century RCP8.5 scenarios. These 95th percentile FWI values capture the extremes that a region may experience with some regularity (i.e., the FWI values do not represent rare, one-off maximums).

Grid cell-level 95th percentile values were used to create classes of relative fire danger across the continental United States, with classes for each decade (e.g., historical FWI class) and for each season (e.g., winter historical FWI class). FWI classes were developed using the distribution of 95th percentile values (for all CONUS grid cells) from the historical period. For example, the "Low" FWI class comprised the 0-25th percentile FWI values for the historical period, rounded to the nearest whole

[4] Canadian Forestry Service. (1984). *Tables for the Canadian Forest Fire Weather Index System* (Forestry Technical Report 25, 4th edition). Environment Canada. https://cfs.nrcan.gc.ca/publications?id=31168

number; the lowest 25th percent of observations in the historical period translated grids with 9 FWI and below. Therefore, all grid cells with values below 9 FWI would be classified as "Low." In full, the classes are:

FWI Class	Percentile range in historical period	FWI values in Class
Low	0–25th percentile	0–9 FWI
Medium	25–50th percentile	9–21 FWI
High	50–75th percentile	21–34 FWI
Very High	75–90th percentile	34–39 FWI
Extreme	90–98th percentile	39–53 FWI
Very Extreme	Above 98th percentile	Above 53 FWI

The same FWI Class thresholds were applied to future (mid-century and end-of-century) FWI projections to aid in visualizing changes in fire danger over time. For example, the "Medium" FWI class encompasses any grid cells with a 95th percentile FWI value from 9 to 21 across all scenarios (historical, mid-century, and end-of-century).

Heat Index

Heat index is calculated using temperature and relative humidity as its two key inputs, with the output measure providing a better picture of how hot the outside weather feels to humans. The heat index information provided in ClimRR is calculated using an extended heat index[5] formula that better captures extreme heat conditions. Using this heat index method is important for understanding how future climate change will produce extremes beyond those typically experienced under historical conditions.

Each climate model used as inputs in ClimRR generates temperature and relative humidity readings every 3 hours, or 8 readings per day. This information was used to calculate heat index for all 8 time points per day. We then selected the maximum heat index for a given day, as it is the extreme conditions for any given day that are of greatest interest.

Since heat index is a measure whose utility primarily applies to hot conditions, rather than cold conditions, we focused our efforts on summer months (June, July, and August). All heat index attributes currently in ClimRR are based on summer measurements. We used the daily maximum heat index readings to calculate several different measures of historical and future heat.

1. **Summer daily maximum heat index**: this output is calculated by taking the average of every day's maximum heat index for the summer months, i.e., about 90 readings per summer. We repeated this for each year within the model period (e.g., 2045, 2046, etc.) and across each climate model (CCSM, GFDL, and HadGEM) to produce the ensemble mean.
2. **Summer seasonal maximum heat index**: this measure finds the single highest maximum heat index reading for a given summer (i.e., the highest daily maximum heat index for 2045). This summer maximum is found for each year in the model period (e.g., 2045, 2046, etc.) and across each climate model (CCSM, GFDL, and HadGEM) to produce the ensemble mean.
3. **Number of summer days with daily max heat index above 95 / 105 / 115 / 125 °F**: these measures use the daily maximum heat index values to calculate the number of summer days in

[5] Lu, Yi-Chuan, and David M. Romps. "Extending the Heat Index." *Journal of Applied Meteorology and Climatology* 61, no. 10 (October 2022): 1367–83. https://doi.org/10.1175/JAMC-D-22-0021.1.

which the heat index exceeds a given threshold value. We use four thresholds in ClimRR: 95°F, 105°F, 115°F, and 125°F. These thresholds represent a ranger of relative heat-related danger to humans. We find the count of the number of day's exceeding the threshold for each year across the modeled period (e.g., for 2045, 2046, etc.) across each climate model (CCSM, GFDL, and HadGEM) to produce the ensemble mean.

ClimRR currently provides the 6 variables detailed above for the historical period (1995-2004), the mid-century period under RCP8.5 (2045-2054), and the end-of-century period under RCP8.5 (2085-2094). For each variable, change characteristics have also been calculated to visualize the growth in heat index measures from historical to mid-century and from historical to end-of-century.

Data Dictionary

This document contains descriptions of all field names within the following climate variables. The links below direct to the rest service of the gridded data.

13. Temperature Minimum – Annual
14. Temperature Minimum – Seasonal
15. Temperature Maximum – Annual
16. Temperature Maximum – Seasonal
17. Precipitation – Annual Total
18. Precipitation None – Annual Average
19. Wind Speed – Annual Average
20. Cooling Degree Days – Annual Total
21. Heating Degree Days – Annual Total
22. Fire Weather Index – Averages
23. Heat Index - Summer

Temperature Minimum – Annual

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist	Min Daily Temperature (°F) – Annual Average – Historical
rcp45_midc	Min Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
rcp45_endc	Min Daily Temperature (°F) – Annual Average – End-Century RCP4.5
rcp85_midc	Min Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
rcp85_endc	Min Daily Temperature (°F) – Annual Average – End-Century RCP8.5
mid45_hist	Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
end45_hist	Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist	Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
end85_hist	Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5

Temperature Minimum – Seasonal

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist_Winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
rcp45_midc_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
rcp45_endc_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
rcp85_midc_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
rcp85_endc_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
mid45_hist_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
end45_hist_winter	Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
end85_hist_winter	Winter - Min Daily Temperature (°F) – Annual Average – Historical
hist_Spring	Spring - Min Daily Temperature (°F) – Annual Average – Historical
rcp45_midc_Spring	Spring - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
rcp45_endc_ Spring	Spring - Min Daily Temperature (°F) – Annual Average – End-Century RCP4.5
rcp85_midc_ Spring	Spring - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
rcp85_endc_ Spring	Spring - Min Daily Temperature (°F) – Annual Average – End-Century RCP8.5
mid45_hist_ Spring	Spring - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
end45_hist_ Spring	Spring - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist_ Spring	Spring - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
end85_hist_ Spring	Spring - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
hist_Summer	Summer - Min Daily Temperature (°F) – Annual Average – Historical
rcp45_midc_Summer	Summer - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
rcp45_endc_Summer	Summer - Min Daily Temperature (°F) – Annual Average – End-Century RCP4.5
rcp85_midc_Summer	Summer - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
rcp85_endc_Summer	Summer - Min Daily Temperature (°F) – Annual Average – End-Century RCP8.5
mid45_hist_Summer	Summer - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
end45_hist_Summer	Summer - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist_Summer	Summer - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
end85_hist_Summer	Summer - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
hist_Autumn	Autumn - Min Daily Temperature (°F) – Annual Average – Historical
rcp45_midc_Autumn	Autumn - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
rcp45_endc_ Autumn	Autumn - Min Daily Temperature (°F) – Annual Average – End-Century RCP4.5
rcp85_midc_ Autumn	Autumn - Min Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
rcp85_endc_ Autumn	Autumn - Min Daily Temperature (°F) – Annual Average – End-Century RCP8.5
mid45_hist_ Autumn	Autumn - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
end45_hist_ Autumn	Autumn - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist_ Autumn	Autumn - Min Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
end85_hist_ Autumn	Autumn - Min Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5

Temperature Maximum - Annual

FIELD NAME	DESCRIPTION
CROSSMODEL	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
HIST	Max Daily Temperature (°F) – Annual Average – Historical
RCP45_MIDC	Max Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
RCP45_ENDC	Max Daily Temperature (°F) – Annual Average – End-Century RCP4.5
RCP85_MIDC	Max Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
RCP85_ENDC	Max Daily Temperature (°F) – Annual Average – End-Century RCP8.5
MID45_HIST	Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
END45_HIST	Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
MID85_HIST	Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
END85_HIST	Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
MID85_45	Max Daily Temperature (°F) – Difference in Annual Average – Mid-Century RCP4.5 and Mid-Century RCP8.5
END85_45	Max Daily Temperature (°F) – Difference in Annual Average – End-Century RCP4.5 and End-Century RCP8.5
MID85_45	Min Daily Temperature (°F) – Difference in Annual Average – Mid-Century RCP4.5 and Mid-Century RCP8.5
END85_45	Min Daily Temperature (°F) – Difference in Annual Average – End-Century RCP4.5 and End-Century RCP8.5

Temperature Maximum – Seasonal

FIELD NAME	DESCRIPTION
CROSSMODEL	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
HIST_WINTER	Winter - Max Daily Temperature (°F) – Annual Average – Historical
RCP45_MIDC_WINTER	Winter - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
RCP45_ENDC_WINTER	Winter - Max Daily Temperature (°F) – Annual Average – End-Century RCP4.5
RCP85_MIDC_WINTER	Winter - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
RCP85_ENDC_WINTER	Winter - Max Daily Temperature (°F) – Annual Average – End-Century RCP8.5
MID45_HIST_WINTER	Winter - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
END45_HIST_WINTER	Winter - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
MID85_HIST_WINTER	Winter - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
END85_HIST_WINTER	Winter - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
HIST_SPRING	Spring - Max Daily Temperature (°F) – Annual Average – Historical
RCP45_MIDC_SPRING	Spring - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
RCP45_ENDC_ SPRING	Spring - Max Daily Temperature (°F) – Annual Average – End-Century RCP4.5
RCP85_MIDC_ SPRING	Spring - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
RCP85_ENDC_ SPRING	Spring - Max Daily Temperature (°F) – Annual Average – End-Century RCP8.5
MID45_HIST_ SPRING	Spring - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
END45_HIST_ SPRING	Spring - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
MID85_HIST_ SPRING	Spring - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
END85_HIST_ SPRING	Spring - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
HIST_SUMMER	Summer - Max Daily Temperature (°F) – Annual Average – Historical
RCP45_MIDC_SUMMER	Summer - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
RCP45_ENDC_SUMMER	Summer - Max Daily Temperature (°F) – Annual Average – End-Century RCP4.5
RCP85_MIDC_SUMMER	Summer - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
RCP85_ENDC_SUMMER	Summer - Max Daily Temperature (°F) – Annual Average – End-Century RCP8.5
MID45_HIST_SUMMER	Summer - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
END45_HIST_SUMMER	Summer - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
MID85_HIST_SUMMER	Summer - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
END85_HIST_SUMMER	Summer - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5
HIST_AUTUMN	Autumn - Max Daily Temperature (°F) – Annual Average – Historical
RCP45_MIDC_AUTUMN	Autumn - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP4.5
RCP45_ENDC_AUTUMN	Autumn - Max Daily Temperature (°F) – Annual Average – End-Century RCP4.5
RCP85_MIDC_AUTUMN	Autumn - Max Daily Temperature (°F) – Annual Average – Mid-Century RCP8.5
RCP85_ENDC_AUTUMN	Autumn - Max Daily Temperature (°F) – Annual Average – End-Century RCP8.5
MID45_HIST_AUTUMN	Autumn - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP4.5
END45_HIST_AUTUMN	Autumn - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP4.5
MID85_HIST_AUTUMN	Autumn - Max Daily Temperature (°F) – Change in Annual Average – Historical and Mid-Century RCP8.5
END85_HIST_AUTUMN	Autumn - Max Daily Temperature (°F) – Change in Annual Average – Historical and End-Century RCP8.5

Precipitation – Annual Total

FIELD NAME	DESCRIPTION
CROSSMODEL	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
HIST	Precipitation (in) – Annual Total – Historical
RCP45_MIDC	Precipitation (in) – Annual Total – Mid-Century RCP4.5
RCP45_ENDC	Precipitation (in) – Annual Total – End-Century RCP4.5
RCP85_MIDC	Precipitation (in) – Annual Total – Mid-Century RCP8.5
RCP85_ENDC	Precipitation (in) – Annual Total – End-Century RCP8.5
MID45_HIST	Precipitation (in) – Change in Annual Total – Historical and Mid-Century RCP4.5
END45_HIST	Precipitation (in) – Change in Annual Total – Historical and End-Century RCP4.5
MID85_HIST	Precipitation (in) – Change in Annual Total – Historical and Mid-Century RCP8.5
END85_HIST	Precipitation (in) – Change in Annual Total – Historical and End-Century RCP8.5
MID85_45	Precipitation (in) – Difference in Annual Total – Mid-Century RCP4.5 and Mid-Century RCP8.5
END85_45	Precipitation (in) – Difference in Annual Total – End-Century RCP4.5 and End-Century RCP8.5

Precipitation None - Annual Average

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist	Consecutive Days with No Precipitation – Annual Ensemble Mean – Historical
rcp45_midc	Consecutive Days with No Precipitation – Annual Ensemble Mean – Mid-Century RCP4.5
rcp45_endc	Consecutive Days with No Precipitation – Annual Ensemble Mean – End-Century RCP4.5
rcp85_midc	Consecutive Days with No Precipitation – Annual Ensemble Mean – Mid-Century RCP8.5
rcp85_endc	Consecutive Days with No Precipitation – Annual Ensemble Mean – End-Century RCP8.5
mid45_hist	
end45_hist	
mid85_hist	
end85_hist	
mid85_45	
end85_45	

Wind Speed – Annual Average

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist	Wind Speed (mph) – Annual Average – Historical
rcp45_midc	Wind Speed (mph) – Annual Average – Mid-Century RCP4.5
rcp45_endc	Wind Speed (mph) – Annual Average – End-Century RCP4.5
rcp85_midc	Wind Speed (mph) – Annual Average – Mid-Century RCP8.5
rcp85_endc	Wind Speed (mph) – Annual Average – End-Century RCP8.5
mid45_hist	Wind Speed (mph) – Change in Annual Average – Historical and Mid-Century RCP4.5
end45_hist	Wind Speed (mph) – Change in Annual Average – Historical and End-Century RCP4.5
mid85_hist	Wind Speed (mph) – Change in Annual Average – Historical and Mid-Century RCP8.5
end85_hist	Wind Speed (mph) – Change in Annual Average – Historical and End-Century RCP8.5
mid85_45	Wind Speed (mph) – Difference in Annual Average – Mid-Century RCP4.5 and Mid-Century RCP8.5
end85_45	Wind Speed (mph) – Difference in Annual Average – End-Century RCP4.5 and End-Century RCP8.5

Cooling Degree Days – Annual Total

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist	Cooling Degree Days – Annual Total – Historical
rcp85_midc	Cooling Degree Days – Annual Total – Mid-Century RCP8.5
mid85_hist	Cooling Degree Days – Change in Annual Total – Historical and Mid-Century RCP8.5

Heating Degree Days – Annual Total

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
hist	Heating Degree Days – Annual Total – Historical
rcp85_midc	Heating Degree Days – Annual Total – Mid-Century RCP8.5
mid85_hist	Heating Degree Days – Change in Annual Total – Historical and Mid-Century RCP8.5

Fire Weather Index - Averages

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
Wildfire_autumn_hist	Seasonal value – Historical
Wildfire_autumn_midc	Seasonal value – Mid-Century RCP 8.5
Wildfire_autumn_endc	Seasonal value – End-Century RCP 8.5
Wildfire_autumn_Dmid	Difference between Mid-Century and Historical
Wildfire_autumn_Dend	Difference between End-Century and Historical
Wildfire_autumn_Pmid	Percent Change between Mid-Century and Historical
Wildfire_autumn_Pend	Percent Change between End-Century and Historical
Wildfire_spring_hist	Seasonal value – Historical
Wildfire_spring_midc	Seasonal value – Mid-Century RCP 8.5
Wildfire_spring_endc	Seasonal value – End-Century RCP 8.5
Wildfire_spring_Dmid	Difference between Mid-Century and Historical
Wildfire_spring_Dend	Difference between End-Century and Historical
Wildfire_spring_Pmid	Percent Change between Mid-Century and Historical
Wildfire_spring_Pend	Percent Change between End-Century and Historical
Wildfire_summer_hist	Seasonal value – Historical
Wildfire_summer_midc	Seasonal value – Mid-Century RCP 8.5
Wildfire_summer_endc	Seasonal value – End-Century RCP 8.5
Wildfire_summer_Dmid	Difference between Mid-Century and Historical
Wildfire_summer_Dend	Difference between End-Century and Historical
Wildfire_summer_Pmid	Percent Change between Mid-Century and Historical
Wildfire_summer_Pend	Percent Change between End-Century and Historical
Wildfire_winter_hist	Seasonal value – Historical
Wildfire_winter_midc	Seasonal value – Mid-Century RCP 8.5
Wildfire_winter_endc	Seasonal value – End-Century RCP 8.5
Wildfire_winter_Dmid	Difference between Mid-Century and Historical
Wildfire_winter_Dend	Difference between End-Century and Historical
Wildfire_winter_Pmid	Percent Change between Mid-Century and Historical
Wildfire_winter_Pend	Percent Change between End-Century and Historical

Heat Index – Summer

Field Name	Description
Crossmodel	Truncated name for "Crossmodel_CellName". Text ID for each cell in the polygon grid.
heatindex_HIS_DayMax	Summer daily max heat index (daily high) -- Historical
heatindex_HIS_SeaMax	Summer seasonal max heat index (seasonal high) -- Historical
heatindex_HIS_Day95	Number of Summer days with daily max heat index above 95 F -- Historical
heatindex_HIS_Day105	Number of Summer days with daily max heat index above 105 F -- Historical
heatindex_HIS_Day115	Number of Summer days with daily max heat index above 115 F -- Historical
heatindex_HIS_Day125	Number of Summer days with daily max heat index above 125 F -- Historical
heatindex_M85_DayMax	Summer daily max heat index (daily high) -- Mid-Century RCP8.5
heatindex_M85_SeaMax	Summer seasonal max heat index (seasonal high) -- Mid-Century RCP8.5
heatindex_M85_Day95	Number of Summer days with daily max heat index above 95 F -- Mid-Century RCP8.5
heatindex_M85_Day105	Number of Summer days with daily max heat index above 105 F -- Mid-Century RCP8.5
heatindex_M85_Day115	Number of Summer days with daily max heat index above 115 F -- Mid-Century RCP8.5
heatindex_M85_Day125	Number of Summer days with daily max heat index above 125 F -- Mid-Century RCP8.5
heatindex_E85_DayMax	Summer daily max heat index (daily high) -- End-Century RCP8.5

heatindex_E85_SeaMax	Summer seasonal max heat index (seasonal high) -- End-Century RCP8.5
heatindex_E85_Day95	Number of Summer days with daily max heat index above 95 F -- End-Century RCP8.5
heatindex_E85_Day105	Number of Summer days with daily max heat index above 105 F -- End-Century RCP8.5
heatindex_E85_Day115	Number of Summer days with daily max heat index above 115 F -- End-Century RCP8.5
heatindex_E85_Day125	Number of Summer days with daily max heat index above 125 F -- End-Century RCP8.5
heatindex_C_M85_DMax	Change in summer daily max heat index -- Historical to Mid-Century RCP8.5
heatindex_C_M85_SMax	Change in summer seasonal max heat index -- Historical to Mid-Century RCP8.5
heatindex_C_M85_D95	Change in number of summer days with daily max heat index above 95 F -- Historical to Mid-Century RCP8.5
heatindex_C_M85_D105	Change in number of summer days with daily max heat index above 105 F -- Historical to Mid-Century RCP8.5
heatindex_C_M85_D115	Change in number of summer days with daily max heat index above 115 F -- Historical to Mid-Century RCP8.5
heatindex_C_M85_D125	Change in number of summer days with daily max heat index above 125 F -- Historical to Mid-Century RCP8.5
heatindex_C_E85_DMax	Change in summer daily max heat index -- Historical to End-Century RCP8.5
heatindex_C_E85_SMax	Change in summer seasonal max heat index -- Historical to End-Century RCP8.5
heatindex_C_E85_D95	Change in number of summer days with daily max heat index above 95 F -- Historical to End-Century RCP8.5
heatindex_C_E85_D105	Change in number of summer days with daily max heat index above 105 F -- Historical to End-Century RCP8.5
heatindex_C_E85_D115	Change in number of summer days with daily max heat index above 115 F -- Historical to End-Century RCP8.5
heatindex_C_E85_D125	Change in number of summer days with daily max heat index above 125 F -- Historical to End-Century RCP8.5